Banksy: Evidence-Gated Repair of Garbled Text in AI-Generated Videos

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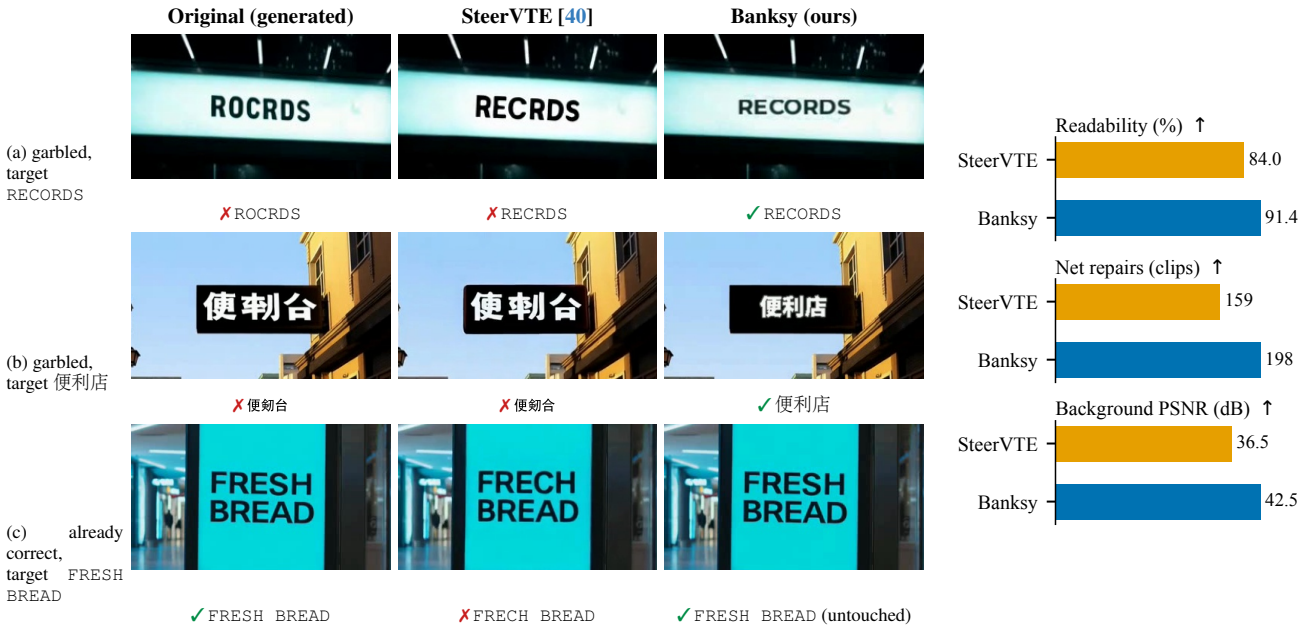


Figure 1. **Repairing garbled text in generated videos.** A generative video text editor rewrites the text region and often produces plausible but wrong characters (a, b) or breaks text that was already correct (c). Banksy edits only when evidence supports it, writes the target string deterministically, and leaves everything outside the text untouched. Below each crop: the transcription of frame 24 by the evaluation reader (✓ correct, ✗ wrong). Right: headline numbers on our 526-clip benchmark (Sec. 5.1). Examples (a, b) were chosen among clips where SteerVTE fails and Banksy succeeds; (c) is the only clip among the 46 that Banksy leaves untouched which SteerVTE breaks.

Abstract

Text-to-video generators still render garbled text on signs, screens and posters. We study repair*: given a generated clip, a target string and a hint box, make the text read correctly and leave everything else as it was. Generative video text editors produce natural glyphs but keep or invent wrong characters, rewrite text that was already correct, and re-render the background. Banksy is a training-free, correctness-first pipeline built around three evidence gates: a preservation gate leaves a clip untouched when multi-frame OCR already reads the target; a localization gate lays the new text out on the old glyph instance when the evidence is strong and falls back to the tracked box otherwise; and an appearance gate enables a non-parametric source appearance transport (SAT) that carries glow and material from the old glyphs to the new ones only where it measurably helps. The target is rendered deterministically, so its spelling cannot drift, and only pixels inside the text region change. On 526 generated clips over seven carrier materials in English and Chinese, Banksy reaches 91.4% readability against 84.0% for the state-of-the-art editor SteerVTE (198 vs. 159 net repairs; a pre-registered second reader confirms the advantage) and keeps the background intact (42.5 vs. 36.5 dB PSNR outside the text). Raters preferred SAT over plain pasting for material fidelity in 87% of non-tied judgements. We report, with pre-registered criteria, what did not work—a learned 1.3B refiner, two appear-*

ance modules, a glyph-locked material generator and reconstruction without any source glyph—and characterise the appearance gap to generative editing that remains.

1. Introduction

Text-to-video models such as Wan [29], CogVideoX [39] and VACE [15] synthesise convincing street scenes, shop fronts and screens, yet the text inside them is often garbled: a record shop reads ROCRDS, a convenience store reads 便刻台 (Fig. 1). In our benchmark of generated clips, only 53.8% of the target strings are rendered correctly. Wherever the text carries meaning, a single wrong character is a hard failure, and re-generating the clip until the text comes out right is expensive and changes the shot. We therefore study *repair*: keep the generated clip, and make its text read correctly.

The natural tool is a generative video text editor, which regenerates the text region conditioned on rendered glyphs and the source style. On generated videos we observe three systematic problems. Such editors write characters that look plausible but are wrong—FRESY BREAD, 小心坍滑—and the strongest of them, SteerVTE [40], still leaves 16% of our clips unreadable. They rewrite text that was already correct (FRESH BREAD → FRECH BREAD, Fig. 1c). And because the whole clip passes through the generator, the background changes too: PSNR outside the text region drops to 36.5 dB.

We argue that repair should be correctness-first and minimal: do not change what is already right; when a change is needed, change only the text; and recover the material of the old text only when there is evidence that this helps. Banksy implements this order with three evidence gates (Sec. 4). A preservation gate returns the clip unchanged when OCR on many frames already reads the target. A localization gate decides whether the new text can be laid out on the old glyph instance—its size, colour and baseline—or must fall back to filling the tracked box. An appearance gate decides whether to apply source appearance transport (SAT), a non-parametric operator that carries glow and material from the old glyphs to the new ones. Between the gates the target is rendered deterministically from a matched font and warped with carrier homographies, so its spelling cannot drift, and only pixels inside the text region are composited. Nothing is trained.

On 526 generated clips covering seven carrier materials in English and Chinese, Banksy raises readability from 53.8% to 91.4% (SteerVTE: 84.0%), with 198 net repairs against 159; the preservation gate does not protect every correct clip, and Banksy breaks 12 of the 283 originally correct ones where SteerVTE breaks 6. It keeps the background intact (42.5 dB). We are equally explicit about what it does not do: its glyphs are rendered, not generated, and where both methods repair a clip, SteerVTE's text is closer to the original appearance in 70% of the cases. We trace this gap to its causes and report the alternatives we tried and rejected under pre-registered criteria.

Our contributions are: **(i)** a repair task, a 526-clip benchmark and a protocol whose metrics separate repairs from breakages and measure background fidelity and temporal stability, with a pre-registered second reader and a blind human study of the appearance step; **(ii)** Banksy, a training-free pipeline in which three evidence gates, each measured for what it is designed to do (breakage, residual old ink at equal readability, appearance), decide whether to edit, where to lay the text out and whether to transport the source appearance, around a deterministic renderer anchored to the tracked carrier; **(iii)** registered negative results—a learned 1.3B refiner, two appearance modules, a glyph-locked material generator and a glyph-free branch that failed its sealed cross-generator test—and a hybrid that accepts SteerVTE's output only where an independent verifier reads the target and otherwise falls back to Banksy, reaching 96.4% readability with 0 broken clips (Sec. 5.6). The full experimental record, including every pre-registration and all audits, is in the supplementary material.

2. Related work

Scene text editing. Early editors decompose a text image into background, style and content and re-synthesise it with GANs [16, 25, 26, 34]; diffusion models brought text rendering and editing to arbitrary images through glyph conditioning [4, 14, 20, 30–32, 38, 41, 46]. Their glyphs are natural, but spelling remains a sampling outcome, and frame-wise application to video flickers. A related line renders text first and harmonises it afterwards [35, 42]; our learned refiner (Sec. 5.5) belongs to it.

Text in generated and edited videos. Text-to-video models render legible text only occasionally [11, 12, 33, 43, 48]; generation-time remedies condition on glyphs [18] or fine-tune on text-rich clips [21] but do not repair a given clip. For editing, SteerVTE [40] injects glyph features and a VLM style embedding into a frozen VACE backbone trained on a million triplets; VidScribe [45] and ViTeX-Bench [5] evaluate video text editing, including commercial models, and find edits that corrupt text that should not change. A concurrent editor [44] warps rendered target glyphs into the text box of every frame as a per-frame reference for a video diffusion model and supervises it with a frozen recognizer, and releases VTEdit, 288 real clips for text replacement and addition; like SteerVTE it regenerates the text region, whereas we place rendered glyphs deterministically on the tracked carrier and gate every edit on evidence. All of these regenerate the text region. We repair generated text with a training-free pipeline that leaves correct text untouched, writes the

target deterministically and changes only the text region, and compare against SteerVTE as the strongest editor.

Text effects, inpainting, evaluation. Example-based text effects transfer keys appearance on the distance to the glyph boundary [24, 36, 37]; SAT fits such an operator on the old glyphs of the same video and gates its use on measured gain. We remove the old text with LaMa [28] once per clip and propagate along the tracked carrier rather than inpainting every frame [3, 47], and we test a carrier-anchored attention bias in the spirit of flow-guided attention [6, 10] on the refiner. Text is scored by OCR [1, 7, 27] and, increasingly, by vision-language models [2]; we use a VLM reader for evaluation and a separate OCR inside the pipeline.

3. Task, benchmark and metrics

Task. The input is a generated clip V, a target string s and a hint box b marking where the text is; in our benchmark s is the quoted sign text of the generation prompt and b comes from OCR localisation and tracking of the generated text, and the same box is given to every method (SteerVTE uses it as its editing mask). The output is a clip $\hat{V}$ of the same size and length in which the text inside b reads s and nothing else has changed. Unlike scene text editing, the source text is a near miss produced by the generator, and in about half of the clips it is already correct, so "do nothing" is often the right answer.

Benchmark. We generated clips of street scenes, shop fronts and screens with VACE-14B [15] from 120 prompts that ask for a specific sign text on one of seven carrier materials, ten seeds each (832×464, 49 frames, 15 fps). From 1,178 usable clips an OCR screen selected the 526 that do not read correctly but still contain a recognisable attempt at the target, and dropped 371 without one; the screen is stricter than our evaluation reader, which judges 53.8% of them correct, so the benchmark contains both repair and do-nothing cases. It therefore measures the repair of a near miss, not reconstruction without any source text, which we study separately (Sec. 5.5). The clips cover 40 target strings (20 English, 20 Chinese) and seven materials: neon (93), signboard (86), paper (96), glass (76), screen (65), metal (56) and wood (54); 295 are English and 231 Chinese. Forty clips inspected during development are listed in the supplement and excluded in a separate report.

3.1. Metrics

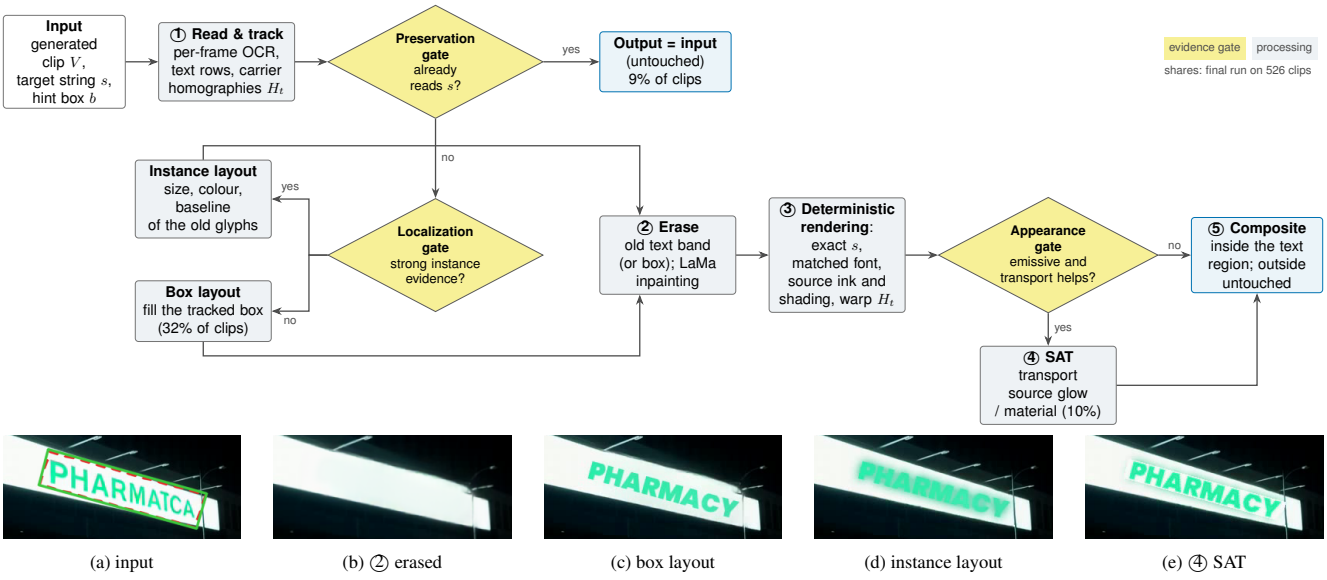
Readability. A vision-language reader, Qwen2.5-VL-7B [2], transcribes the text inside the hint box on 5 frames; a clip is *correct* if at least 4 transcriptions match the target exactly (after normalising whitespace, case and traditional/simplified Chinese). The reader is independent of Banksy, whose gates use a different OCR. All compared outputs are read in one job and one software environment, and a second reader from another model family re-reads the main comparison (Sec. 5.1). **Repairs and breakages.** We count *fixed* (garbled → correct) and *broken* (correct → wrong) clips and their difference, the *net* repairs. **Locality.** *BG PSNR* is computed outside the dilated hint box over all frames; *collateral change* is the grey-level change inside the text neighbourhood but outside both the old and the new glyphs (it needs the method's glyph mask, so it is reported for Banksy only). **Appearance.** On three frames we compare, in CIELAB, the ink colour, ink–ground contrast, halo, edge softness and stroke shading of each output with those of the original glyphs, each normalised by its median over the evaluation run; lower is closer, values are comparable only within one run, and the unseeded ink segmentation gives a test–retest noise of 0.47 per clip, so we interpret only differences well above it. **Temporal stability.** With the carrier motion estimated once on the original and applied to every output, *flicker* is the motion-compensated grey-level change inside the box, *jitter* the same for gradient magnitudes, and *boundary jump* the change at the chunk boundaries of a chunked model. **Human judgements** come from blind pairwise studies with the original shown between two versions in random order and separate questions per quality (supplement). Full definitions are in the supplementary material, Sec. A.

4. Banksy

Banksy takes a clip $V = \{I_t\}_{t=1}^{T}$, a target s and a hint box b and returns a clip that differs from V only inside the text region (Fig. 2). All components are classical or pre-trained; nothing is trained for the task. Thresholds and implementation details are in the supplementary material, Sec. A.

Reading the text and tracking its carrier. An off-the-shelf OCR system (EasyOCR [13]: CRAFT [1], CRNN [27]) runs on every frame; boxes overlapping b are grouped into rows, and the frame whose rows best match s becomes the reference frame $t^\star$. The text sits on a physical *carrier*—a sign, a poster, a screen—so we track the carrier rather than the glyphs: SIFT keypoints [22] around the reference quad are matched to every frame and a RANSAC homography H_t [9] is estimated, with the quad-to-quad homography of the OCR boxes as fallback. Everything placed on the carrier later—erased background, glyphs, masks—is placed once in $I_{t^\star}$ and carried to frame t by H_t, which removes the per-frame jitter of OCR boxes from the output.

Preservation gate. Generated text is often already correct, and an edit can only make such a clip worse. The gate returns $\hat{V} = V$ when the pipeline's OCR reads the com-



(a) input (b) ② erased (c) box layout (d) instance layout (e) ④ SAT

Figure 2. **Banksy overview.** Bottom: intermediate results of the frozen pipeline on an illustrative benchmark clip (garbled `PHARMATCA`, target `PHARMACY`), cropped from one reference frame: (a) input with the OCR box (dashed red) and the accepted glyph instance (green); (b) the band erased with LaMa; (c) the box layout used as fallback; (d) the instance layout, rendered deterministically; (e) after SAT. Three evidence gates decide how much to change: the preservation gate keeps a clip whose text already reads the target, the localization gate chooses between the old glyph instance and the tracked box, and the appearance gate enables SAT where it measurably helps. The target is rendered deterministically and composited only inside the text region. All results attributed to Banksy (B1) are produced by this training-free pipeline; hybrid and exploratory variants are evaluated separately.

plete target in at least 85% of the frames, inside the hint box, with wrong readings in at most 5% of the frames and with the concatenated reading of all boxes inside the box equal to s (which prevents skipping when the target appears next to other, garbled text). It never consults the evaluation reader, and it trades recall for safety: in the final run it skips 46 of the 526 clips, all confirmed correct by the independent reader.

Localization gate and layout. If the old glyphs can be localised reliably, the new text inherits their size, colour and baseline. We estimate a glyph *instance* in the reference frame—ink mask, ink and ground colours, letter height, number of letters, letter band—and accept it only with at least three letters (two for Chinese), an ink–ground contrast above 40 in CIELAB, a letter height between 0.3 and 1.0 of the OCR box height and a regular row grouping. Otherwise the clip takes the *box layout*, which fills the tracked box with legible text; in the final run, 168 of the 480 edited clips (35%) do. With an accepted instance the target is laid out on the letter band at the measured letter height, in the string's natural aspect ratio, centred on the old text; stacked text keeps its rows.

Erasure and deterministic rendering. The old glyphs are removed once, in the reference frame: the ink mask (extended by the bright pixels of glowing text) is dilated, flat grounds are pre-filled with their median colour, and the hole is inpainted with LaMa [28]; the erased background is carried to every frame by H_t with a per-frame gain fitted on a ring around the hole, so exposure changes follow without inpainting flicker. The font is chosen from a library of 33 Latin and 8 Chinese families by comparing stroke statistics of the warped old ink mask with each candidate rendered in the same layout, and the target is drawn by a font rasteriser, so its spelling is exact by construction. Its colour follows the per-frame ink colour of the old text modulated by a low-frequency shading field estimated from the old strokes; glowing text receives a soft halo; the canvas is warped to every frame with H_t and composited over the erased background.

Appearance gate and source appearance transport. On emissive carriers—neon, lightboxes, screens—the old glyphs carry glow, bloom and inner structure that a flat fill cannot reproduce. SAT transports this appearance from the old glyphs to the new ones without regenerating the region. Let B be the erased background, I the reference frame and $d(x)$ the signed distance of pixel x to the boundary of the old glyphs in units of the letter height. On the old text we fit, per distance bin k and per channel, a multiplicative-plus-

additive light-transport operator

$$I(x) \approx m_k\, B(x) + a_k, \qquad k = k(d(x)), \tag{1}$$

over the support $d \le 0.6$, and apply it to the new glyphs with their own distance field, $\hat{I}(x) = m_{k(\hat{d}(x))} B(x) + a_{k(\hat{d}(x))}$. On the old text this family reconstructs the source better than an additive-only operator, a distance-indexed replacement or a flat fill (cross-validated CIELAB error 17.2 vs. 25.9, 27.6 and 29.7; supplement). SAT is applied only behind an *appearance gate*: the text must be emissive and the operator must generalise across the text—fitted on one half and tested on the other, in both directions, with an error below 0.75 of the flat-fill error. In the final run SAT is enabled on 50 clips (10%), never on box-layout clips.

Compositing. The output differs from the input only where the erasure mask or the new glyphs (with halo) are non-zero; elsewhere the input pixels are copied, so the only residual outside the text region is the re-encoding of the video.

5. Experiments

Protocol. Table 1 and every number in Secs. 5.1, 5.2 and 5.5 come from one front-end pass over the 526 clips with the final system and its registered variants, read together with the original clips and SteerVTE's outputs in one evaluation job (Sec. 3.1). Every other table comes from its own pass and its own evaluation run; because the appearance distance is normalised within a run, it is compared only within a table. Paired differences carry 95% bootstrap CIs over clips (10,000 resamples) [8] and readability differences exact McNemar tests [23]. Table 1 reports all clips and, separately, the 486 clips never inspected during development. **Baseline.** SteerVTE [40], the strongest published video text editor (glyph and style conditioning on a frozen VACE-14B backbone, one million training triplets), is run with the hint box as editing mask and the target as text at the input's resolution and length.

5.1. Main results

Banksy makes 91.4% of the clips readable, 7.4 points more than SteerVTE (70 clips correct only with Banksy, 31 only with SteerVTE; $p = 1.3 \times 10^{-4}$), and repairs 210 of the 243 garbled clips. It breaks 12 of the 283 originally correct clips where SteerVTE breaks 6, so its advantage in net repairs (198 vs. 159) is smaller than in repairs alone; the picture is unchanged outside the development set (91.2% vs. 84.4%). The two methods trade different qualities: Banksy leaves the background intact (PSNR 6.0 dB higher, 95% CI [5.8, 6.3]) and its strokes jitter less (7.86 vs. 8.30), because every glyph is placed once on the tracked carrier; SteerVTE's region flickers less and, above all, its text looks more like the original (appearance distance 1.83 vs. 2.57), a gap we analyse in Sec. 6. Banksy is ahead on all seven materials, most on signboards (+15.1 points) and neon (+8.6), least on metal (+1.8) and wood (+1.9), and reaches 91.5% on English and 91.3% on Chinese clips (SteerVTE: 84.1%, 84.0%). Figure 3 shows random examples, including two failures of Banksy.

A second reader. Readability depends on the reader, so we registered a re-reading of the outputs by readers from other model families with identical crops, frames, prompt, decoding and scoring, and a rule fixed in advance: the advantage counts as reader-robust only if InternVL2.5-8B finds Banksy more readable than SteerVTE with $p < 0.05$. It does: 84.8% against 78.7% (74 vs. 42 discordant clips, $p = 3.8 \times 10^{-3}$), with a smaller margin that is significant for Chinese ($p = 0.02$) but not for English alone. A third reader, Pixtral-12B, cannot read Chinese and finds no advantage on the English clips (77.6% vs. 80.3%); most of its errors on Banksy's outputs are correct transcriptions followed by invented text, which exact matching counts as wrong. Full tables are in the supplement, Sec. B.

Edit locality. SteerVTE's difference to the original covers the entire frame, including edges far from the text; Banksy's is confined to the text region, with a residual of 1–2 grey levels elsewhere from re-encoding. Inside the text neighbourhood, the collateral change outside both old and new glyphs is 1.68 grey levels on the 480 clips Banksy edits.

5.2. The evidence gates

Preservation gate. The gate skips 46 clips, all confirmed correct by the reader (no false skip; the registered limit was 2%). In the selective-editing pass (Sec. 5.4) it is the only difference between B0 and B1: it restores 2 of the 16 originally correct clips that B0 breaks and loses none. Its recall is low by design: the pipeline's OCR rarely reads stylised originals—neon italics, extruded or condensed letters—consistently enough, and all 12 clips Banksy breaks are such cases (Sec. 6).

Carrier anchoring, font matching, localization gate. Pasting the target into the per-frame OCR box makes the text readable but nearly doubles flicker (3.12 vs. 1.98 for the original), because OCR boxes jitter; anchoring everything on the tracked carrier brings it to the level of SteerVTE (1.67 vs. 1.66) with lower jitter (7.12 vs. 8.30) and unchanged readability. Matching the font to the old strokes adds 2.6 points of readability ($p = 0.044$) once candidates are rendered in the layout of the original. Laying the text out on the glyph instance leaves half as much of the old ink visible as the box layout (residual 0.21 vs. 0.42) but, with a

Table 1. **Main results** (526 clips; one front-end pass and one evaluation job for every row). Readability: share of clips whose text the reader transcribes exactly in at least 4 of 5 frames. Fixed / broken: garbled → correct and correct → wrong transitions relative to the original as read in the same job; net = fixed − broken. Appearance: glyph-agnostic material distance to the original text (normalised within this run; lower is closer). BG PSNR: outside the dilated text region. Flicker and jitter: motion-compensated, inside the text box (grey levels). The lower block excludes the 40 clips inspected while developing the selective-editing components.

	Readability ↑	Fixed ↑	Broken ↓	Net ↑	NED sim. ↑	Appearance ↓	BG PSNR ↑	Flicker ↓	Jitter ↓
All 526 clips									
Original (generated)	53.8	–	–	–	0.820	–	–	1.98	9.18
SteerVTE [40]	84.0	165	**6**	159	0.946	**1.83**	36.49	**1.66**	8.30
Banksy	**91.4**	**210**	12	**198**	**0.959**	2.57	**42.52**	1.79	**7.86**
486 clips outside the development set									
Original (generated)	54.7	–	–	–	0.822	–	–	1.97	9.12
SteerVTE [40]	84.4	150	**6**	144	0.945	**1.85**	36.40	**1.65**	8.24
Banksy	**91.2**	**189**	12	**177**	**0.958**	2.60	**42.50**	1.79	**7.86**

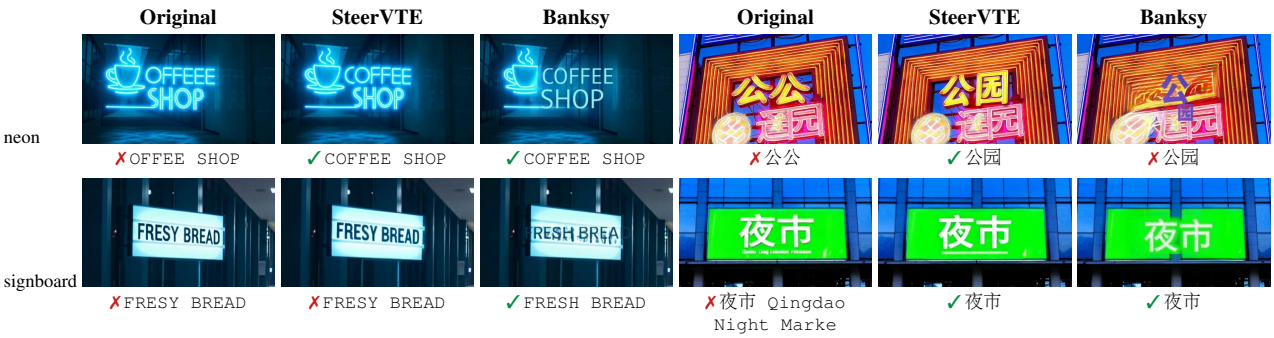


Figure 3. **Qualitative comparison on randomly sampled clips** (two of the four materials of the full figure in the supplement, Sec. B) (one English and one Chinese clip per material, drawn with a fixed seed among garbled originals outside the development set; not curated). Each cell is the same crop of frame 24; below it, the reader's transcription (✓ correct, ✗ wrong). SteerVTE produces more natural glyphs but frequently keeps or invents wrong characters; Banksy writes the target but its glyphs often look pasted (see Sec. 6).

confidence-based tracker, cost 5.5 points of readability; carrier tracking recovers most of it, and the localization gate routes the unreliable instances to the box layout, giving a front end as readable as the box layout (91.1 vs. 90.7) with the instance path's lower residual and lowest jitter. Two readings of the same outputs differ by about one point, so we claim parity, not superiority (supplement, Sec. B).

5.3. Source appearance transport

The appearance gate enables SAT on 53 clips of the analysed pass (23 neon, 11 screen, 9 signboard, 8 metal, 2 wood); most rejections are clips without an accepted instance, non-emissive text, or no cross-validated gain. On the enabled clips SAT reduces the appearance distance by a third (−1.02; better on 42 of 53 clips) for every material—screens 3.02→1.33, neon 3.03→2.02, wood 4.12→2.21—without hurting readability or the background, at a cost in temporal stability (flicker +0.15, jitter +0.81). Because the distance is only a proxy, we ran a blind study on all 53 clips: raters saw the original between the paste-only and the SAT version in random order and answered three separate questions. With four raters (one excluded by a rule fixed after seeing the data; results with all five are reported too), SAT was judged closer to the original material in 158 of 182 non-tied judgements (win rate 0.87, 95% CI [0.77, 0.94]; 48 vs. 3 clips by majority) and as having fewer artifacts in 147 of 191 (0.77 [0.66, 0.87]). The measured flicker increase was not perceived. The six clips raters judged worse show a bright rectangular plate behind the new letters, which we traced to the canvas truncating the halo's distance field; the fix was diagnosed on these clips and is therefore not part of the evaluated system (supplement, Sec. B and F).

5.4. Selective, structure-preserving editing

The most frequent problem of an early Banksy was not missing material but changing things it should not have. We therefore tested, with criteria fixed before the run, three components added to B0 (Banksy without the preservation gate): the preservation gate (B1), glyph-only erasure (B2) and a carrier-fit check that declares a target layout-infeasible (B3). In one front-end pass and one evaluation run, the preservation gate meets its criteria (no false

skip, two fewer broken clips, readability not lower) and is adopted, which makes B1 the Banksy of Tab. 1. Glyph-only erasure reduces collateral change on all clips (-0.090, 95% CI $[-0.16, -0.02]$) but not significantly outside the development clips, and costs 7.6 points of readability because the instance ink mask is not a complete glyph mask; of 24 clips flagged infeasible, inspection confirmed 5. B2 and B3 are rejected (supplement, Sec. B).

5.5. What did not pass

We report the alternatives we built and rejected; each was judged by criteria fixed before its run, and all details are in the supplement, Sec. C and D.

A learned refiner. A video diffusion transformer initialised from Wan2.1-T2V-1.3B [29], trained as a flow-matching bridge [17, 19] from the pasted clip to the target on Blender renders, 562 real pairs and up to 1.15 million image pairs, does not improve on the paste it starts from: refiners trained with 0, 10k, 100k and 1M image pairs reach 75–78% readability on a garbled subset against 80.7% for the paste, and a diagnostic on 1,500 held-out pairs shows that they move away from the original even on the task they were trained for, because the paste's font and the generator's glyphs make the supervision inconsistent. Glyph-consistent supervision recovers most of the image-domain loss but still does not exceed the paste on video. On the full benchmark the best refiner costs 3.5 points of readability, adds flicker, jitter and chunk seams, and leaves the appearance distance unchanged (-0.035, 95% CI $[-0.21, 0.19]$). The result is specific to our scale (1.3B parameters, a few hundred real video pairs).

Two appearance modules. Baseline-aligned layout with glyph-matched fonts (B1f) changes the output in 157 and 25 clips respectively and leaves readability unchanged, but its appearance difference ($+0.052$ [$+0.002$, $+0.106$]) fails the registered criterion; the appearance distance is glyph-agnostic, so it cannot see what B1f changes, and we report the criterion unchanged rather than amend it. Reuse of the correctly generated source glyphs (R1), which keeps the right letters in place with their own material and renders only the missing ones, applies to only 7 of the 526 clips with our current segmentation, matching and gating, too few to evaluate. Neither module is part of Banksy.

A glyph-locked material generator. A conditional U-Net in the carrier's UV space, trained on 2,000 Blender counterfactual pairs that differ only in the text, closes 0.48 of the gap between a flat paste and the rendered truth on held-out pairs (registered gate 0.50; a model with 1.8× the parameters, the one registered capacity test, closes 0.498). Applied zero-shot to 129 benchmark clips it fails every registered criterion (appearance distance 3.06 vs. 1.99 for Banksy; 20 readings lost). Under our registered protocol, increasing capacity did not close the synthetic-domain criterion and zero-shot transfer to generated video degraded both appearance and readability; a photometric ablation attributes part of the transfer loss to the renders' photometry and leaves the rest unexplained by it.

Table 2. **Verified selection between Banksy and SteerVTE** (526 clips, one evaluation run; the full table with ablations is in the supplement, Sec. B). Banksy-V keeps the clip when the preservation gate fires, otherwise takes a SteerVTE output whose text an independent verifier (InternVL2.5-8B) reads as the target, composited locally into the original, and falls back to Banksy. W/L: clips correct only with the system / only with Banksy.

	Read. ↑	Broken ↓	Net ↑	W/L	Appear. ↓	BG PSNR ↑
Banksy (B1)	91.8	12	201	–	3.34	42.52
SteerVTE	84.0	5	160	29/70	2.37	36.49
Banksy-V (verified, local)	96.4	0	225	30/6	2.67	41.74
Oracle selection (upper bound)	98.1	0	234	34/1	2.70	41.74

Reconstruction without source glyphs. Banksy's gating assumes that some glyph survives; every benchmark clip has at least pseudo-text, which its detection-box fallback reads at 90.1% (SteerVTE: 76.1%). We additionally investigated glyph-free, carrier-conditioned text reconstruction. On a sealed cross-generator evaluation (CogVideoX-5B and Wan2.2-TI2V-5B, 265 glyph-erased clips), its frozen final version improved readability by 20.0 percentage points over the static-hint fallback (95% CI +14.8 to +25.5). However, it did not satisfy all acceptance criteria registered before the test set was opened, with failures in text-scale consistency and temporal non-inferiority. We therefore retain the evidence-gated B1 system as our primary method and report glyph-free reconstruction as an exploratory extension (supplement, Sec. D).

5.6. Verified selection between Banksy and SteerVTE

The two systems fail on different clips, so a system that could tell which output is right would inherit the readability of both. We registered one before running it, with no tunable threshold: *Banksy-V* keeps the clip when the preservation gate fires; otherwise it runs SteerVTE (seed 1, then a second seed) and accepts an output whose text an independent verifier—InternVL2.5-8B with the reading protocol of Sec. 3.1—transcribes as the target in at least 4 of 5 frames, composited back into the original around the text box with a feathered boundary and a colour match; otherwise it falls back to Banksy. The verifier never scores. In one run over every row (Tab. 2), Banksy-V reads correctly in 96.4% of the clips (30 vs. 6 discordant against Banksy, $p = 7.0 \times 10^{-5}$) with 0 broken clips and 225 net repairs; its

appearance distance lies between SteerVTE's and Banksy's, because SteerVTE's text is used wherever it reads correctly, and the background stays local (41.74 dB). It meets five of six registered criteria and misses the temporal one by a margin that we trace to the re-encoding of the composited output (supplement, Sec. B). Banksy-V is a system combination—its material from SteerVTE, its correctness from verification, its locality and fallback from Banksy—and it runs SteerVTE up to twice per clip plus a 7B verifier; a registered blind study of its naturalness against SteerVTE is in progress and is not reported here.

5.7. External benchmarks

On SteerVTE's own VTE-Bench Real (100 real clips whose correct text is to be changed), the frozen Banksy is clearly behind SteerVTE (73.0% vs. 92.0%), partly because its front end does not locate 8 of the source texts and ships those clips unchanged: on correct, well-formed text the gates that make Banksy safe for repair have nothing to protect. On the 29 replacement and translation clips of VidScribe [45] that OCR can locate, Banksy reads 37.9%, ahead of the strongest of seven commercial models (31.0%) without significance and significantly ahead of 3 of them, with the background kept (36.2 dB against 16.7–28.0 dB). We report both to delimit the claim: Banksy is a repair method for garbled generated text, not a general video text editor (supplement, Sec. B).

6. Analysis and limitations

Where Banksy fails. Banksy breaks 12 of the 283 originally correct clips (SteerVTE: 6). In all of them the evaluation reader transcribes the original correctly while the pipeline's OCR reads it in at most 2% of the frames (neon italics, extruded or condensed letters), so the preservation gate cannot fire; the outputs then show new text larger than its sign (4 clips), old text left visible next to the new text (5), wrong colour or contrast on lightboxes (2) or a misread stacked text (1). The more common failure does not affect readability: erasure leaves a visible patch, or the new glyphs are flat where the old ones were outlined, extruded or glowing. These cases explain the appearance distance in Tab. 1.

The appearance gap to generative editing. Among the 145 garbled clips that both methods repair, SteerVTE's text is closer to the original in 70%, so the gap is not an artefact of SteerVTE leaving garbled text untouched. Four causes, in decreasing order of visibility: Banksy renders instead of generating, so it cannot reproduce extrusion, bevels, tube geometry or texture inside the strokes; its font library is small (3 Chinese families are ever selected, and Noto Sans SC alone renders 67% of the edited Chinese clips); its layout inherits the axis-aligned orientation of the OCR quad, so on tilted carriers the text can look skewed; and LaMa's fill is visible as a patch on textured carriers. Two observations suggest that the gap is closable without giving up correctness: the garbled source already contains a median of 88% of the target's letters (Chinese: 67%), so reusing them would extend "change only what is wrong" from words to characters once letter segmentation is reliable; and the two methods fail on different clips (at least one repairs 97.3%), which the verified selection of Sec. 5.6 exploits.

Other limitations. The benchmark comes from one family of generators and asks for short sign texts; long texts are out of scope. Readability is measured by VLM readers, which favour large, high-contrast text and can transcribe a garbled original as the intended word; the second reader confirms the advantage over SteerVTE with a smaller margin, and the third finds the methods level on English once its invented additions are discounted, so we report all of them. The appearance distance is glyph-agnostic and noisy at the clip level; it supports large differences such as the gap to SteerVTE and the effect of SAT, not small ones, and it rewards a constant colour, which is why a flat paste scores well on it. The SAT study used four raters; the blind comparison of Banksy-V with SteerVTE is in progress. The frozen front end is bit-exact within one software environment but not across OpenCV and CUDA versions, where its font matcher and a few OCR decisions diverge (about 3 readability points and 1.4 dB of background PSNR on 71 clips); every comparison in this paper is therefore made within one environment and one evaluation run (supplement, Sec. G). Outside repair, on editing correct text, the frozen system is clearly behind SteerVTE (Sec. 5.7).

7. Conclusion

Repairing garbled text in generated videos is not the same problem as editing text: the source is a near miss, it is often already right, and everything around it should survive. Banksy treats repair as a sequence of evidence-gated decisions—leave the clip alone, lay the text out on the old glyphs or on the tracked box, transport the old appearance or not—around a rendering step whose spelling cannot drift. This makes it more reliable than the strongest generative editor on our benchmark and confines its changes to the text, but its glyphs still look rendered. Our negative results delimit what closing that gap requires: an appearance measure that sees glyph shape and orientation, letter segmentation reliable enough to reuse the characters the generator drew correctly, and generative edits accepted only when they can be verified, as verified selection does at the cost of running the generative editor.

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LLM usage statement

Large language models were used as assistants for writing and editing the text of this paper, for drafting experiment scripts, and for summarising evaluation outputs. The task definition, the method, the pre-registered criteria, the experiments and their interpretation are the authors' own; every number in the paper is generated from the evaluation files of the stated runs, and the authors verified the claims against those files. Vision-language models are also used as components of the evaluation (readers) and of the verified-selection system, as described in the paper.